

Warmup

Group members:

Instructions: Work with a neighbor on the following activity. I will collect the handout at the end of class, and it will be part of your class participation grade. You will be graded only on effort – it is ok if you don’t finish all the questions, or get them all correct.

Titanic data

Consider the fitted model for the Titanic data:

Consider the model

$$Survived_i \sim Bernoulli(\pi_i)$$

$$\log \left(\frac{\hat{\pi}_i}{1 - \hat{\pi}_i} \right) = 3.78 - 0.037Age_i - 2.52Male_i - 1.31Class2_i - 2.58Class3_i$$

1. What is the estimated *probability* of survival for a male, first-class passenger aged 20?
What about for a male, second-class passenger aged 30?

2. For each of these two passengers, do you predict they will survive or not? (That is, make a binary prediction – a “yes” or “no” answer – for each passenger)